Mintong Kang

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Research Profile

I develop safety and security methods for advanced AI systems, with a focus on automatic red-teaming, robust guardrails, and certifiable safety guarantees for LLMs, multimodal models, audio-language models, and agentic AI systems. My work spans both attacks and defenses: uncovering emerging risks in frontier models and building reliable mechanisms to mitigate them through reasoning, policy grounding, and formal certification.

Education

- **University of Illinois Urbana-Champaign** — **Urbana-Champaign, IL, USA**
  Ph.D. in Computer Science — *Aug 2022 – Present*
  - Advisor: Prof. Bo Li.
  - Research areas: AI safety and security, LLM agents, multimodal red-teaming, trustworthy machine learning, and certifiable safety.
  - GPA: 3.91/4.0.

- **Zhejiang University** — **Hangzhou, China**
  Bachelor of Computer Science and Technology — *Aug 2018 – Jun 2022*
  - Advisor: Prof. Xi Li.
  - GPA: 3.95/4.0.
  - Honors: Outstanding Undergraduate Award, Outstanding Thesis Award, and First-Prize Scholarship.

Publications

(* stands for equal contribution) (first- and co-first-author publications are highlighted)

Representative themes: **AI safety evaluation and red-teaming**; **guardrails and policy reasoning**; **certifiable trustworthy ML**; **fair and robust generation**.

20. Zhaorun Chen, Xun Liu, **Mintong Kang**, Jiawei Zhang, Minzhou Pan, Shuang Yang, Bo Li. ARMs: Adaptive Red-Teaming Agent against Multimodal Models with Plug-and-Play Attacks. *Fourteenth International Conference on Learning Representations (ICLR)*, 2026.
19. **Mintong Kang**, Zhaorun Chen, Bo Li. C-SafeGen: Certified Safe LLM Generation with Claim-Based Streaming Guardrails. *Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
18. **Mintong Kang***, Zhaorun Chen*, Chejian Xu*, Jiawei Zhang*, Chengquan Guo*, Minzhou Pan, Ivan Revilla, Yu Sun, Bo Li. Poly-Guard: Massive Multi-Domain Safety Policy-Grounded Guardrail Dataset. *Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
17. Chejian Xu, **Mintong Kang**, Jiawei Zhang, Zeyi Liao, Lingbo Mo, Mengqi Yuan, Huan Sun, Bo Li. AdvAgent: Controllable Blackbox Red-teaming on Web Agents. *Forty-Second International Conference on Machine Learning (ICML)*, 2025.
16. Zhaorun Chen, **Mintong Kang**, Bo Li. ShieldAgent: Shielding Agents via Verifiable Safety Policy Reasoning. *Forty-Second International Conference on Machine Learning (ICML)*, 2025.
15. **Mintong Kang**, Bo Li. R^2-Guard: Robust Reasoning Enabled LLM Guardrail via Knowledge-Enhanced Logical Reasoning. *Thirteenth International Conference on Learning Representations (ICLR)*, 2025. **Spotlight.**
14. **Mintong Kang**, Chejian Xu, Bo Li. AdvWave: Stealthy Adversarial Jailbreak Attack against Large Audio-Language Models. *Thirteenth International Conference on Learning Representations (ICLR)*, 2025.

13. **Mintong Kang**, Vinayshekhar Bannihatti Kumar, Shamik Roy, Abhishek Kumar, Sopan Khosla, Balakrishnan Murali Narayanaswamy, Rashmi Gangadharaiah. FairGen: Controlling Sensitive Attributes for Fair Generations in Diffusion Models via Adaptive Latent Guidance. *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.
12. Lingbo Mo*, Zeyi Liao*, Chejian Xu, **Mintong Kang**, Jiawei Zhang, Chaowei Xiao, Bo Li, Huan Sun. EIA: Environmental Injection Attack on Generalist Web Agents for Privacy Leakage. *Thirteenth International Conference on Learning Representations (ICLR)*, 2025.
11. Chejian Xu*, Jiawei Zhang*, Zhaorun Chen*, Chulin Xie*, **Mintong Kang***, Zhuowen Yuan*, Zidi Xiong*, Chenhui Zhang, Lingzhi Yuan, Yi Zeng, Peiyang Xu, Chengquan Guo, Andy Zhou, Jeffrey Ziwei Tan, Zhen Xiang, Zinan Lin, Dan Hendrycks, Dawn Song, Bo Li. MMDT: Decoding the Trustworthiness and Safety of Multimodal Foundation Models. *Thirteenth International Conference on Learning Representations (ICLR)*, 2025.
10. **Mintong Kang**, Nezihe Merve Gürel, Ning Yu, Dawn Song, Bo Li. C-RAG: Certified Generation Risks for Retrieval-Augmented Language Models. *Forty-First International Conference on Machine Learning (ICML)*, 2024.
9. **Mintong Kang**, Zhen Lin, Jimeng Sun, Cao Xiao, Bo Li. Rob-FCP: Certifiably Byzantine-Robust Federated Conformal Prediction. *Forty-First International Conference on Machine Learning (ICML)*, 2024.
8. **Mintong Kang***, Nezihe Merve Gürel*, Linyi Li, Bo Li. COLEP: Certifiably Robust Learning-Reasoning Conformal Prediction via Probabilistic Circuits. *Twelfth International Conference on Learning Representations (ICLR)*, 2024.
7. Boxin Wang*, Weixin Chen*, Hengzhi Pei*, Chulin Xie*, **Mintong Kang***, Chenhui Zhang*, Chejian Xu, Zidi Xiong, Ritik Dutta, Rylan Schaeffer, Sang T. Truong, Simran Arora, Mantas Mazeika, Dan Hendrycks, Zinan Lin, Yu Cheng, Sanmi Koyejo, Dawn Song, Bo Li. DecodingTrust: A Comprehensive Assessment of Trustworthiness in GPT Models. *Thirty-Seventh Conference on Neural Information Processing Systems (NeurIPS)*, 2023. **Outstanding Paper Award**.
6. **Mintong Kang**, Dawn Song, Bo Li. DiffAttack: Evasion Attacks Against Diffusion-Based Adversarial Purification. *Thirty-Seventh Conference on Neural Information Processing Systems (NeurIPS)*, 2023.
5. **Mintong Kang**, Linyi Li, Bo Li. FaShapley: Fast and Approximated Shapley Based Model Pruning Towards Certifiably Robust DNNs. *IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*, 2023.
4. **Mintong Kang**, Yongyi Lu, Alan L. Yuille, Zongwei Zhou. Data, Assemble: Leveraging Multiple Datasets with Heterogeneous and Partial Labels. *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2023.
3. **Mintong Kang***, Linyi Li*, Maurice Weber, Yang Liu, Ce Zhang, Bo Li. Certifying Some Distributional Fairness with Subpopulation Decomposition. *Thirty-Sixth Conference on Neural Information Processing Systems (NeurIPS)*, 2022. **Spotlight**.
2. Bhaskar Ray Chaudhury, Linyi Li, **Mintong Kang**, Bo Li, Ruta Mehta. Fairness in Federated Learning via Core-Stability. *Thirty-Sixth Conference on Neural Information Processing Systems (NeurIPS)*, 2022. **Spotlight**.
1. Hanbin Zhao, Yongjian Fu, **Mintong Kang**, Qi Tian, Fei Wu, Xi Li. MgSvF: Multi-Grained Slow vs. Fast Framework for Few-Shot Class-Incremental Learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2021.

Preprints

3. Zhaorun Chen, Xun Liu, Haibo Tong, Chengquan Guo, Yuzhou Nie, Jiawei Zhang, **Mintong Kang**, Chejian Xu, Qichang Liu, Xiaogeng Liu, Tianneng Shi, Chaowei Xiao, Sanmi Koyejo, Percy Liang, Wenbo Guo, Dawn Song, Bo Li. DecodingTrust-Agent Platform (DTap): A Controllable and Interactive Red-Teaming Platform for AI Agents. *arXiv preprint*, 2026.
2. **Mintong Kang**, Chen Fang, Bo Li. AudioGuard: Toward Comprehensive Audio Safety Protection Across Diverse Threat Models. *arXiv preprint*, 2026.
1. **Mintong Kang**, Chong Xiang, Sanjay Kariyappa, Chaowei Xiao, Bo Li, Edward Suh. Mitigating Indirect Prompt Injection via Instruction-Following Intent Analysis. *arXiv preprint*, 2025.

Industry Experience

- **Google DeepMind** — **New York, NY, USA**
  Student Researcher, Gemini Audio Post-training Team — *Feb 2026 – Present*
  - Evaluating and improving the function-calling capability, safety, and security of Gemini Live audio agents.

- **Virtue AI** **Remote**
  Part-time Research Intern, advised by Prof. Bo Li *2025 – 2026*
  - Developed automatic red-teaming methods to evaluate safety and security risks in advanced LLMs, reasoning models, multimodal systems, and agentic AI.
  - Collaborated with frontier AI labs (e.g., **OpenAI**, **Google DeepMind**) and industry research teams on internal red-teaming of reasoning models, scientific agents, and text-to-image generation systems.
- **NVIDIA Research** **Santa Clara, CA, USA**
  Research Intern, advised by Dr. Edward Suh *May 2025 – Dec 2025*
  - Developed reasoning-time defenses for LLM agents against indirect prompt injection by analyzing instruction-following intent signals in model reasoning traces.
  - Translated intent analysis into policy-aware defensive decisions for tool-using agents, achieving state-of-the-art performance on agentic prompt-injection defense benchmarks.
  - Filed a U.S. patent and collaborated with the NVIDIA Nemotron team on practical deployment of the defense algorithm.
- **Amazon AWS** **Santa Clara, CA, USA**
  Research Intern, advised by Dr. Rashmi Gangadharaiah *May 2024 – Dec 2024*
  - Studied fairness evaluation and bias mitigation for text-to-image diffusion models across sensitive attributes and high-impact domains.
  - Developed adaptive latent guidance for controlled fair generation, leading to *FairGen*, accepted to EMNLP 2025.

Selected Awards

- **NeurIPS 2023 Outstanding Paper Award** 2023
- **Cybersecurity Award: Best Machine Learning and Security Paper** 2024
- NeurIPS Scholar Award 2023
- NeurIPS Scholar Award 2022
- Outstanding Undergraduate Award, Zhejiang University 2022
- Outstanding Undergraduate Thesis Award, Zhejiang University 2022
- First-Prize Scholarship, Zhejiang University 2021, 2020
- First Prize, Chinese Olympic Mathematics Competition League 2018, 2017

Selected Talks

Evaluation and Red Teaming of LLM Agents, **Google DeepMind** Apr 2026
Evaluation and Red Teaming of LLM Agents, **Capital One Research** Mar 2026
Certifiable Safety and Security for Advanced AI Systems, **Berkeley Security Seminar** Sep 2025
Instruction-Following Intent Analysis for Prompt-Injection Defense, **NVIDIA Research** Aug 2025
Certifiable Autonomous Space Driving, **NASA University Leadership Initiative** Jun 2025
Adaptive Red Teaming for LLM Agents, **AI Times** Feb 2025
Certified Safe Generation and Trustworthy LLM Guardrails, **University of Georgia Seminar** Sep 2024
Certifiable Trustworthy Machine Learning for Foundation Models, **UIUC Machine Learning Seminar** Sep 2024
Trustworthy AI: From Robustness and Fairness to LLM Safety, **AI Times** Feb 2024

Services

- **Reviewer**: ICML 2022, 2023, 2024, 2025, 2026
- **Reviewer**: NeurIPS 2022, 2023, 2024, 2025, 2026
- **Reviewer**: ICLR 2024, 2025, 2026
- **Reviewer**: TMLR 2026
- **Reviewer**: AISTATS 2025
- **Reviewer**: CVPR 2023, 2024, 2025
- **PC Member**: AAAI 2024, 2025
- **Workshop Organizer**: ICML 2023 Workshop on Knowledge and Logical Reasoning in the Era of Data-Driven Learning
- **Competition Organizer**: NeurIPS 2024 Competition on LLM and Agent Safety
- **Teaching Assistant**: UIUC CS 441: Applied Machine Learning, Spring 2025

Selected Research Artifacts

- GitHub **DecodingTrust**: Codebase for comprehensive trustworthiness evaluation of GPT models across safety, robustness, fairness, privacy, ethics, and reliability dimensions.
- GitHub **DecodingTrust-Agent**: Interactive red-teaming framework for AI agents, supporting controllable environments, agent trajectory collection, and safety evaluation.
- GitHub **DiffAttack**: Implementation of evasion attacks against diffusion-based adversarial purification, including attack pipelines and evaluation scripts for robust vision models.
- Hugging Face **DecodingTrust Dataset**: Benchmark dataset accompanying DecodingTrust for evaluating trustworthiness risks in large language models.
- Hugging Face **DTap-Bench Agent Trajectories**: Agent trajectory dataset for studying failures, risks, and red-teaming scenarios in tool-using AI agents.
- Hugging Face **PolyGuard**: Large-scale multi-domain, policy-grounded guardrail dataset covering safety rules and moderation scenarios across social media, finance, law, education, HR, code, cyber, and AI regulation.